Measure results (fill in)

Write each number with its unit. Photo: the file name of the caliper photo in `originals/measure_2026-11/`. Items are explained in MEASURE_LIST.md.

A. Wiper motor (params.py `WIPER`, set `measured=True` when done)

Item	Result (with unit)	Photo file or note .....
A1 Output shaft diameter		
A2 Shaft flats or spline		
A3 Shaft end thread and length		
A4 Mount hole pattern		
A5 Housing size		
A6 Free speed and free current		
A7 Stall torque (optional)		
A8 Mounting boss and hub details		

B. Launch springs (params.py `SPRING`, set `measured=True` when done)

Item	Result (with unit)	Photo file or note .....
B1 Free length L0		
B2 Rate k and initial tension F0		
B3 Wire and outside diameter		
B4 Which preload hole		

C. Parts box: confirm you have these (no new purchases)

Item	Result (with unit)	Photo file or note .....
C1 608 bearings (buy 10)		
C2 One M8 x 25 bolt and nut (the CAD's bought list), and any M8 ...		
C3 Worm N20 count: you need 4 (dart kick, deploy, foot, and the ...		
C4 MP1584 #2 (the spare from the dart build; confirm it, decision ...		
C5 DRV8833 #3 (freed from the dart by the new pin map)		
C6 100 kg load cell + HX711 (August order)		
C7 Resistors: 2 x 10 k, 1 x 1 k, 1 x 680 R, 1 x 100 k, 1 x 22 k		
C8 Spare XT60 plugs, 14 AWG wire, two small inline fuse holders ...		
C9 Barrel plug 5.5 x 2.5 mm with lead		
C10 Yellow RCA lead		
C11 USB data cable for the DevKitC, STEMMA QT to Dupont cable		
C12 M3 screws (lengths from the base print list), M3 nuts, M3 ...		
C13 AS5600 magnet (recommended): any small neodymium disc (6 to 10 ...		
C14 Launch springs: two only (decision D38)		
C15 8 mm steel rod (or M8 bolts) for: reel shaft S, aim worm wheel ...		
C16 Hardware the CAD assumes: M3 x 4 x 5 heat-set inserts; M3 ...		
C17 JST SH 4-pin (STEMMA QT) cable 400 mm or longer for the AS5600		
C18 Carriage stop		
C19 No PETG needed: the brake rocker prints in PPA-CF with an ...		

Item	Result (with unit)	Photo file or note .....
C20 How many 693ZZ / F693ZZ bearings are left after the dart (the ...		
C21 Coupon extras, exactly as the coupon sheets list them ...		

D. Electrical labels to read

Item	Result (with unit)	Photo file or note .....
D1 RC832 input voltage		
D2 Jetson DC input range		
D3 DRV8833 carrier make		
D4 DevKitC board version		
D5 Video standard		

E. Rover

Item	Result (with unit)	Photo file or note .....
E1 Deck size and mounting holes		
E2 Drive train and whether the Jetson drives it		
E3 Kiyo X height and tilt on the rover		
E5 Rover mass, wheelbase, centre of mass		
E4 Base launch height		

F. Dart (from the KEDGE notes, still open)

Item	Result (with unit)	Photo file or note .....
F1 Dart v30 is the one being built		
F2 Bought part weights and the finished dart weight		
F3 Stall current of each dart motor at 2S		
F4 OTOTEC BMS size and F693ZZ flange		
F5 Nose ballast		

G. Bench and field numbers (measured during the TEST_PLAN stages)

Item	Result (with unit)	Photo file or note .....
G1 BTS7960 current sense scale		
G2 Battery voltage reading		
G3 AS5600 zero and drum positions		
G4 Storage reel payout drag		
G5 Muzzle speed		
G6 Range at three pitches		
G9 Line tension per amp (dart-side tension, decision D43)		
G10 Tow limit (choose)		
G11 Payout brake trim		
G12 Contact gaps		
G13 AS5600 air gap		
G14 Wiper coupler shear pin (before the first powered cock, ...		
G7 Soil holding force		
G8 Rail friction		

H. Other parts the CAD assumed

Item	Result (with unit)	Photo file or note .....
H1 Drum N20 output shaft		
H2 Drum N20 stall torque at 9 V		

Item	Result (with unit)	Photo file or note .....
H3 CSK20PP keyway depths		
H4 BTS7960 module		
H5 DevKitC USB socket		

After you measure

Item	Result (with unit)	Photo file or note .....